Erik Hill

AI Evaluation & Testing Engineer

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SUMMARY

Self-taught AI evaluation and testing engineer building deterministic, replayable systems for measuring coding agents and LLM applications.

I build **verifiable-evaluation systems** whose claims are checked rather than asserted: **a public registry of 12 accepted evaluation claims, each hash-pinned to a commit and replayed weekly in CI** (11 of 12 replayed green on 2026-10-19): deterministic, no LLM judges, failures published by design. Code merged upstream at promptfoo and TeaVM; a game shipped to Google Play; **30+ public repos, three PyPI packages, one npm package.**

SKILLS

Evals & AI: Evals · No-LLM-judge grading · Mutation testing · Differential-oracle testing · Property-based testing · Regression gating (nDCG) · Pre-registration & minimum-detectable-effect analysis · RAG · LLM application development · Tool-calling & guardrails · Safety (OWASP LLM Top 10, human-in-the-loop) · Prompt engineering · Context engineering · Model Context Protocol (MCP) · LangGraph · Multi-agent orchestration
Backend & data: Python · FastAPI · Pydantic · SQLAlchemy 2.0 · PostgreSQL · pgvector · Alembic · REST / OpenAPI
Frontend / mobile: TypeScript · JavaScript · React · Next.js 16 · Java · Capacitor · Android (adb) · WebAssembly (Pyodide) · TeaVM (JVM to JavaScript)
Ship & verify: pytest · Vitest · JUnit · jqwik · Playwright · PyPI / npm packaging · GitHub Actions (matrix CI) · Docker · Cloudflare Pages (Wrangler) · Neon · Test automation · QA for LLM systems

SELECTED PROJECTS, all public at github.com/egnaro9

• **agent-certlab**: **Two real coding agents certified, every verdict regradeable from artifacts**. A certification lab built on three **seeded-defect task families** with provable answer keys (incl. **coordinated two-file defects** no single-file patch can pass), a **precondition prover** (clean passes, every seed fails, it refused one of my own defects), and **calibration agents** the harness must separate before certifying.

• **reference-fleet**: **An answer key for eval suites**: six models each broken **one documented way at an exactly reproducible seeded rate**, so detection rates are measurable against ground truth, a naive suite detects **1 of 6** (paired protocol). Plus a **LoRA-trained native defect**: mixture 0.507 → realized **0.20 greedy, 0.31 at temp 1.0**: training's rate control is lossy, measured.

• **vac-protocol**: **Verifiable Agent Claims**: an evidence-bundle format + **offline verifier**; a registry where **12 accepted claims are hash-checked and replayed weekly in CI** (11 of 12 replayed green on 2026-10-19), with a **public record of it rejecting my own evidence**. Plus **vac-gate**: replayable evidence or no green check.

• **evalmut**: **Mutation testing for eval graders** (**PyPI**, with its **gradecore** engine): inject known defects, report which checks stayed green. Found **4 holes in the graders of gradecore**, the library it depends on, and **8 in ports of a widely-used framework's assertions**. **No LLM judge**; every operator tied to a documented real failure. Fixed **three fail-open graders** in gradecore 0.10.3, each reproduced against shipped code first. *325 tests.*

• **deadgate**: **a linter for CI checks that cannot fail** (**PyPI**). Reads GitHub Actions workflows for gates a failure cannot reach. Across **275 repositories** I hand-labelled every one of its 653 findings and **deleted two of its four detectors** because hand-labelled evidence said they had never once been right: **36.1% false, measured rather than estimated**. Found an unguarded required status check in **heygen-com/hyperframes**. *228 tests.*

• **deadcheck**: **does an eval rubric catch what it claims?** Run on **three rubrics I did not write**: OpenAI evals' `Match` accepts "10" for ideal "1"; a promptfoo example passes refusal-then-compliance. Seven probe rounds found **23 defects under its own green suite**. *185 tests.*

• **rag-eval-lab**: deterministic RAG **eval harness** that catches a planted hallucination in CI; from-scratch **BM25 matching the published SciFact baseline (0.664 vs 0.665).** *60 tests.*

• **agent-graph**: a **LangGraph ReAct agent** whose multi-step tool use is **unit-testable**: a swappable mock/LLM policy so the graph runs deterministically without a key, a **safe AST calculator** instead of eval, and a max-step guard. *67 tests.*

• **model-drift**: public board, **frozen hash-fingerprinted suite** across **17 models / 5 labs**, with a derived **minimum-detectable-regression floor**.

• **harness-builder**: **More scaffolding scored worse, for 22x the cost**. Three harness shapes on one 20-task suite, **zero errors in 60 runs**: one drafter 95%, planner plus drafter 90%, four-call panel with judge 80%. The panel won **0 of 20 tasks**, and the tool reported **"this suite cannot decide"** rather than ranking.

• **tapdodge-engine**: **One rulebook, two runtimes, and a check that fails when they stop agreeing**. A JVM-versus-browser (TeaVM) differential fails CI when the two disagree. In a committed fault injection, the differential was the **only** check that caught a runtime-supplied-RNG fault: the same-runtime determinism test and the golden file both passed it.

• **Also public: exposure-watch** (AI agents vs. plain code at known-exploited-vulnerability triage; on a pre-registered held-out set of 57 pairs, a **tie on correct answers**, 47 to 45, p = 0.69, but the skeptic-checked agent pipeline gave **0 wrong answers** to 12 from code plus a keyword rule) · **crashkit** (live) · **mcp-tools** (MCP server, no SDK) · **pi-gates** (fail-closed commit and model gates for the pi coding agent) · **pi-eval** (no-LLM-judge grading for pi, on gradecore) · **llm-wire-stub** (npm) · **readme-test-count** (fails CI when a README's advertised test count disagrees with the suite; runs in the CI of six of my repos)

REVIEWED BY OTHERS

• **Two pull requests merged into promptfoo** (26k stars), both merged by its co-founder and CTO. #11278: a blank BLEU reference, such as an unset template variable, let a one-word answer score 1 against a longer reference; the fix shipped in promptfoo 0.124.0. #10256: the `rouge-l` / `rouge-s` assertion types, merged 2026-10-09.

• **External audit of vac-protocol:** an outside engineer falsified my own verifier, cheapest forgery **four bytes**. Fix merged, then I found the same defect class **four more times** (arXiv, below). **Sole maintainer through the audit**: merged four of his pull requests and answered his four specification issues in SPEC.md (protocol v0.2). On two I showed the fix he proposed would not close the hole and shipped a wider one, reasoning public.

• **Peer review with Bartłomiej Nawara**, senior AI/ML engineer: his missing-not-at-random critique of my rate-limit handling shaped **model-drift**'s **reliability floor**, credited in the repo. I found **two design faults** in his pre-registered study.

• **Closed a bug report open in TeaVM since 2018** that its own suite could not catch: a **one-character error** in TeaVM's GregorianCalendar emulation, found by differential-testing the browser port against the JVM. The four assertions covering that line had been commented out; the patch re-enables them. Merged same day by the maintainer.

PUBLICATIONS

"Four Ways to Forge a Bundle My Own Verifier Calls Clean: Refusal-Site Mutation Testing of an Evidence-Bundle Verifier." Erik Hill. **arXiv preprint, 2026**. arXiv:2608.26183
25+ articles on evaluation and testing failures, **100+ comments from 25+ readers**. dev.to/agentdev9 · **Drift Notes**: erikhill.dev/notes

EXPERIENCE

AI/ML Engineer · Epix Devo LLC · *Contract · Oct 2026 to Present*

Independent AI Evaluation & Testing Engineer · Solo Practice · *Remote · Mar 2026 to Present*

- Designed **five roles across a four-stage production loop** (Strategy → Execution → Critic → Ops) that plans, builds, reviews, and validates changes to a real React / Capacitor product, human-in-the-loop on every consequential step, plus an **asynchronous Evaluation harness** that scores completed batches.
- Built per-role model routing with manager-driven handoffs: each stage runs as a fresh subagent pinned to the model its role needs; enforced rules with **property-based invariants** (jqwik) over a native engine and **adb-driven on-device validation**. Case study: exposure-watch/docs/HARNESS.md
- **Operate model-drift as a running service, not a demo**: its scheduled probe has run on every calendar day since 2026-07-19 and publishes a public board.
- Diagnosed an apparent **66-point overnight drop** on that board (69% to 3% on a frozen suite) as a **provider rate cap rather than model drift**: 34 of 35 calls hit a 429 against a 30 RPM shared tier. Shipped Retry-After-aware retries, a per-host throttle, and a reliability floor that drops low-reliability probes from the accuracy line, so a rate limit can no longer render as a regression.

Indie Game Developer · SeraphLight Studios · *Remote · Nov 2025 to Present*

- **Shipped Tap Dodge Rush to Google Play** (Kotlin / Java, Android Views + Canvas, AdMob), closed testing with 12+ testers, then store review, then launch. A second, larger game (React / Capacitor) is in development, the product the harness above operates on.

Kitchen Manager / Shift Lead · Big Island Brewhaus & other high-volume kitchens · *Hawaii · 2018 to 2025*

- Led shifts and trained staff; owned prep, inventory, and food safety under sustained pressure.

EDUCATION

Clark College, 2007 to 2009 (some college, no degree).